

MRINMOY KUMAR MAITY

AI/ML Engineer | Generative AI & Python Developer

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SUMMARY

Entry-level AI/ML Engineer building practical Python applications with LLMs, RAG, semantic search, and REST APIs. Developed and deployed full-stack AI systems using FastAPI, React, Groq, Ollama, and Pydantic, with experience in structured outputs, local embeddings, retrieval workflows, and automated testing.

SKILLS

Programming: Python, SQL, JavaScript

AI/ML: Machine Learning, Generative AI, LLMs, RAG, Embeddings, Semantic Search, Prompt Engineering, LangChain, LangGraph

Backend & Web: FastAPI, Flask, REST APIs, Pydantic, React, Vite, Streamlit

Data & Analytics: Pandas, NumPy, scikit-learn, Power BI, Tableau, Excel, Matplotlib, EDA

Tools & Databases: Git, GitHub, MySQL, Ollama, Wireshark

EXPERIENCE

Deloitte Australia Data Analytics Virtual Experience Program

May 2025 - Jul 2025

Forage | Remote

- Analyzed business and forensic datasets to identify inconsistencies, patterns, and actionable findings.
- Built an interactive Tableau dashboard and used Excel classification to communicate results in a structured business report.

PROJECTS

Candidate AI Representative

Aug 2026

Python, FastAPI, React, Vite, Groq API, Pydantic | [Live Demo](#) | [GitHub](#)

- Built and deployed a full-stack, resume-grounded AI portfolio assistant with streaming responses and Pydantic-validated profile data.
- Exposed 9 REST endpoints for candidate Q&A, PDF resume upload/download, interview-question generation, job-description matching, and recruiter-facing summaries.
- Added speech-to-text, text-to-speech, English/Hindi/Bengali support, conversation history, and PDF chat export; deployed on Vercel and Render.

Vector Search & Local RAG System

Jul 2026

Python, Flask, Ollama, nomic-embed-text, Llama 3.2 | [GitHub](#)

- Built a semantic search and RAG system with document chunking, local embeddings, context-aware Q&A, and a Flask REST API.
- Implemented brute-force, KD-tree, and HNSW-style search across cosine, Euclidean, and Manhattan distance; added 2D visualization and latency comparison.

Deep Packet Inspection & Network Traffic Analyzer

Aug 2026

Python, TCP/IP, PCAP, TLS, HTTP, DNS, Wireshark, JSON | [GitHub](#)

- Reimplemented an open-source C++ packet analyzer in pure Python to parse classic PCAP files across Ethernet, VLAN, IPv4, TCP, and UDP.
- Extracted TLS SNI, HTTP Host, and DNS domains; tracked bidirectional flows; classified applications; filtered by IP, app, or domain; generated PCAP and JSON output.
- Wrote automated tests, including an end-to-end 77-packet capture that verified 71 correctly forwarded packets.

ResumeParser AI - Candidate Screening System

Jul 2026

Python, Groq LLM, Pydantic, PDFPlumber, python-docx | [GitHub](#)

- Parsed PDF/DOCX resumes and produced validated JSON, weighted job-match scores, rankings, and missing-skill explanations.

EDUCATION

Bachelor of Computer Applications (BCA)

Expected Nov 2027

Kharagpur College, West Bengal | Specialization: Computer Science

Higher Secondary Education (Class XII)

2023

Jhargram Kumud Kumari Institution, West Bengal | Computer Science

CERTIFICATIONS

- Introduction to Agent Skills & Claude 101 - Anthropic (Jul 2026); Make Agentic AI Work for You - IBM (Jun 2026)
- Data Analysis Using Python - IBM (Aug 2025); GenAI in Data Analytics - Meta (Jul 2025)